

“Ensemble model of CARD”

CovidNet, Attention ResNet, DenseNet

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Abstract—Enhancing the accuracy of X-ray image classification is an essential factor to prompt treatment. This is the main purpose that we study three conventional neural networks, DenseNet, CovidNet, and Attention ResNet to investigate their performances in classifying the chest X-ray images into three classes, Normal, Covid-19, and pneumonia. Finally, we use weighted ensemble model to have all the advantages of these three networks together. The result of this study was classifying the images with an accuracy of 95.85%.

Index Terms—Attention-based network, Residual network, DenseNet, classification, Optimization, Neural Networks.

I. INTRODUCTION

The main objective of medical image classification is to identify (kind of) disease with high accuracy to assist medical specialists. These classification based on CNN has achieved good results and it is still worth to try in increasing the accuracy because of the importance of the result in people's lives. Specifically in the classification of chest X-ray images, diagnosing the disease and starting the treatment in the early stage is essential.

Since CNNs have proven as a strong classifier for medical images, in this project, we try to implement and study the performance of three networks: DenseNet, CovidNet, and Attention ResNet. Then using an ensemble model enhance the total performance. The accuracy of applying the ensemble model on the test dataset is 95.85%.

DenseNet as an architecture that overcomes problems like gradient vanishing in deep neural networks tries to save maximum information flow between layers. This network strengthens feature propagation and reduces the number of parameters [1]. Therefore it can play a crucial role in image classification. Of all the models, this one has the best accuracy with an acceptable training time.

CovidNet as a network specifically designed for Covid-19 detection has a light-weight network architecture, which consists of both a microarchitecture and macroarchitecture design [2]. To decrease the computational complexity, the network only leverages long range connections in special parts of the structure [3].

In the Attention ResNet, we propose implementing a convolutional neural network using an attention mechanism that can use the feed-forward network architecture in an end-to-end training fashion. [4]

This network consists of some attention modules that cause top information to be gathered sequentially and decides where to attend for the next feature learning steps.

In this module, we have two branches, the mask branch is responsible for adding soft weights on features. It helps the network to insist on more important features.

After a comparison between the three models' performances, we investigate the potential benefits of the ensemble model in increasing accuracy. In ensemble learning, we use more than one models, that are combined together to produce a single output. We are combining the predictions of various models to have the final prediction and based on this prediction we will do the classification. In this work, we multiply prediction to a weight that is defined by the minimum accuracy of all models [5].

The main contributions of this paper are:

- Analysing and comparison between DenseNet, CovidNet, and Attention ResNet.
- Investigating the effect of the ensemble model in image classification.
- Evaluation of the models and finding the weakness and strengths of each model for future research.

The next sections of this paper are as follows:

section II is a brief review of the related work.

section III consists of an explanation about the high-level pipeline of the whole project and the main structure of each model.

section IV is a description of data and its preprocessing.

section V has the detail of all model structures and training processes.

section VI is a discussion about the result and findings.

section VII is a summary of all the project results.

II. RELATED WORK

DenseNet architecture is one of the powerful convolutional neural networks and shows state-of-art in different areas. The most important part of it is the connections between layers. Unlike the traditional feed-forward networks in DenseNet, there are direct connections from any layer to all subsequent layers. In L layers DenseNet, l_{th} layer has l inputs (that they concatenate as one input) but in traditional networks, it has just one input from the previous layer. Therefore in the DenseNet with L layers there are $L(L+1)/2$ connections that help to bring the information from the early layers to the late one [1].

[6], [7], [8] use the DenseNet for extracting features from the chest X-ray images. In [7] and [6] they use DenseNet as a network for extracting the 1024 features from X-ray images

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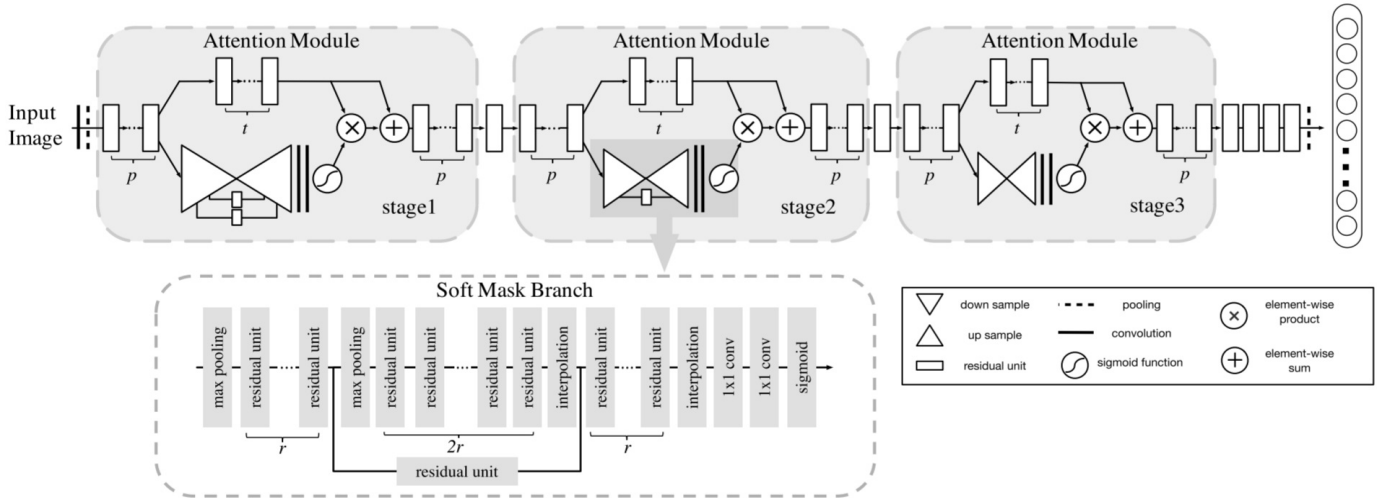


Fig. 1: The architecture of Attention ResNet. In this project we put $p = 1$, $t = 2$, $r = 1$

and try to give the score for the extent of lung involvement and degree of opacity. In [8] they use DenseNet to distinguish between the 14 different pathologies.

[3] introduced a network that can detect the covid-19 cases from chest X-ray images. This network, CovidNet, has an outstanding performance compared to other networks. They have several updates in their networks to enhance the performance [9] that one of the best ones is explained here [2].

The attention mechanism in Neural Networks tends to mimic the cognitive attention that human being has. This mechanism utilizes top information to guide the bottom-up feedforward process. Recently, some studies have been done on applying attention to deep neural networks. One of them is [10] which presents a novel recurrent neural network model for extracting information from an image or video. In this paper, they use sequential decision methods. There is another interesting study [4]. They use Stacked Attention Modules that gradually refine the feature maps. Their model achieved 0.6% top-1 accuracy improvement.

III. PROCESSING PIPELINE

DenseNet There are two main blocks in this network: Dense and transition blocks. In the Dense block, we have dense connections which are direct connections from any layer to all subsequent layers. The dense connection causes each layer receives the concatenation of feature-maps of all previous layers. Therefore each layer has access to all the preceding feature-maps in its block. To concatenate feature-maps, they should have the same size. hence, in dense block sizes of the input and output are the same. The transition block, as a bridge between the dense blocks is responsible to change the feature-maps size to make them suitable for the next block.

CovidNet One of the networks we use in this project is Covid-Net CXR-2 and the main block is Project-replication-projection-expansion (PRPE). The main block has a pattern with low computational complexities and higher representational performance. In this block, input features are replicated through two pointwise convolutions, then output features with depthwise convolution are split and finally, they are mixed via two pointwise convolutions. In the network, we have two types of PRPE blocks: PRPE, and PRPE-S, the second one has the stride = 2 in depthwise convolution to reduce spatial dimensionality more.

In Covid-Net CXR-2 we use long-range connections when it is necessary. Unlike densely connected deep neural network architectures, there are four central hubs of long-range connectivity, that are responsible to connect earlier layers to much later layers. These hubs are densely connected 1×1 convolution layers.

Attention ResNet The Residual Attention Network is a combination of multiple Attention Modules. Each Attention Module has two branches: the mask branch and the trunk branch. The trunk branch does the feature processing and the mask branch contains fast feed-forward sweep and top-down feedback steps. The first part collects global information about the image. The output of this bottom-up feedforward structure is low-resolution feature maps with strong semantic information. Then the top-down network combines global information with original feature maps and the Skip connection is between the bottom and top feature maps to capture information from different scales. The output of the mask branch $M(x)$ softly weight output of the trunk branch $T(x)$.

Figure 1 can show the process of this network.

Wighted Ensemble When we have different models, we can use the ensemble to use the strong part of each model and have all the strengths together. To enhance the overall accuracy

of image classification in this project, we use a weighted ensemble model. In this model, we should combine the result of our networks to increase the final accuracy of classification. In combining the performances, we use weighted summation. The weights are defined by using all networks' validation accuracy.

IV. SIGNALS AND FEATURES

A. Dataset

For this project, we use chest X-ray images in three classes: covid, pneumonia, and normal. The total number of samples is 4575 images and there are 1525 images in each class. All images are in RGB color space. We got our data from this source [11] and split our dataset to 80% for train data and 20% for test data. From the train part, we took 20% for the validation dataset. The number of each set is :
train dataset = 2928 images, validation dataset = 732 images, test dataset = 915 images.

B. Preprocessing

we resized all images to 224×224 then normalized by sample mean and standard deviation. Labels of classes in datasets were defined by integers, we convert them into one-hot.

In the next step, we should do the augmentation just on our train dataset to increase the diversity of our training set. For this purpose, we use ImageDataGenerator of Keras with some parameters that we can see below:

random rotation: It changes the angles at that objects appear in the dataset during training. We chose 25 for the degree range in random rotation.

width and height shift: It shifts randomly the images, either towards the left or right (up or down). We chose 0.25 as the upper bound of the fraction of the total width(height) by which the image is to be randomly shifted.

zoom: This parameter zoom in or zoom out the image. We chose 0.2 which is less than 1 and magnifies the images.

horizontal flip: randomly images flipped horizontally.

brightness: It changes the brightness of images randomly and in this project the value of shift is in the range of 0.8 to 1.2.

V. LEARNING FRAMEWORK

A. DenseNet

As we mentioned in the section III we have two main blocks. we will explain them below:

1) *Dense block*: : The main core of this block is a composition of three sequential operations: batch normalization, Rectified linear unit (ReLU), and a 3×3 convolution. we call this core "Conv". The Dense block consists of different numbers of Convs as it is shown in figure 2.

This block is called Denes as it has dense connections. The dense connection means a direct connection from any Conv to all subsequence Convs. The output of one Conv

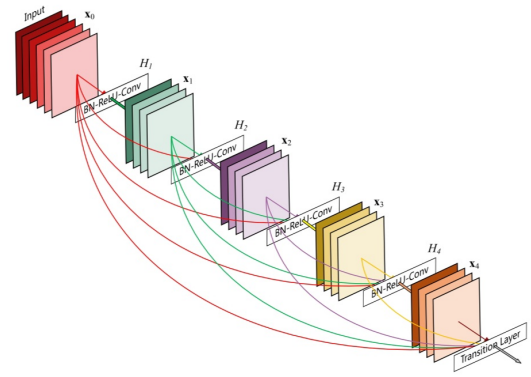


Fig. 2: The Dense block

concatenate with all previous Convs and passed to the next Convs. It causes maximum information flow between layers in the network. Concatenating information learned by different Convs enhances variation in the input of subsequent Conv and improves efficiency. This block has a hyperparameter growth rate K , to define the number of output feature-map after each Conv. It shows the number of feature-maps that are added to the flow of information in the dense block after concatenation. K is the filters parameter in the 3×3 convolution layer in Conv.

2) *Transition block*: : Concatenating required that all the learned feature-map be of the same size. At the same time, the essential part of CNN is to reduce feature-map size. It is not possible to densely connect all Convs when the size of its output feature-map is decreased. the coordination between these two aims is done by transition blocks. The network is divided into dense blocks which are connected by transition blocks together. The transition block consists of Conv, 2×2 average pooling with stride = 2. The average pooling is responsible for halving size.

In this work, we use DenseNet-BC, B refers to bottleneck layers and C refers to compression.

The bottleneck layers consist of a Conv with 1×1 convolution layer to reduce numbers of input feature-maps and a Conv with 3×3 convolution layer, and the growth rate of 1×1 convolution is $4k$.

compression is for improving the compactness of the model by halving the number of the filters in the Convs of transition blocks.

we implemented the DenseNet-BC with 121 layers and $k=32$ (DenseNet-121, $k=32$). All details are shown in table 3.

To train the model we use Adam optimizer. We set the initial learning rate to 0.001. The learning rate is multiplied to 0.9 each time the validation accuracy plateaus after 5 epochs.

B. CovidNet

As we mentioned in section III there are two main blocks PRPE and PRPE-S. They consist of sequential operations: 1×1 convolution, 1×1 convolution, 3×3 depth-wise convolution, 1×1 convolution, 1×1 convolution.

There are four main paths in our architecture. Each path has one 1×1 convolution as a "hub". In the first path, we

Layers	Output Size	DenseNet-121($k = 32$)
Convolution	112×112	7×7 conv, stride 2
Pooling	56×56	3×3 max pool, stride 2
Dense Block (1)	56×56	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$
Transition Layer (1)	56×56	1×1 conv
	28×28	2×2 average pool, stride 2
Dense Block (2)	28×28	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$
Transition Layer (2)	28×28	1×1 conv
	14×14	2×2 average pool, stride 2
Dense Block (3)	14×14	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 24$
Transition Layer (3)	14×14	1×1 conv
	7×7	2×2 average pool, stride 2
Dense Block (4)	7×7	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 16$
Classification Layer	1×1	7×7 global average pool
		fully-connected, softmax

Fig. 3: The DenseNet architecture

have three blocks of PRPE. In the second path, the first block is PRPE-S and three PRPE blocks. The next two paths are like the second path, but the number of PRPE blocks are respectively 5 and 2. In each path, all blocks are densely connected. it means the input of each block is the concatenate of all previous blocks.

The output of each path (concatenate of all blocks and hub of the path) goes to the first block and also to hub of the next path.

This architecture is shown in figure 4

To train the model we use Adam optimizer. We set the initial learning rate to 0.0002. Same as the DenseNet network, the learning rate is multiplied to 0.9 each time the validation accuracy plateaus after 5 epochs.

C. Attention ResNet

In the implemented network we have three attention modules, which are connected by residual units. There is the possibility to increase the number of modules due to increasing performance.

1) *residual units*: In this work, a pre-activation Residual Unit is used. In this unit, three times we have the sequential batch normalization, ReLU, and convolution layers. These three blocks are the same except for filters, kernel size, and strides of the convolution layers. filters will be set in the network, and respectively kernel sizes are 1,3,1, and strides are 1,2,1. After three blocks the input of the residual unit adds element-wisely to the output of the last convolution.

2) *Attention module*: In each module, we have two branches: the trunk branch and the mask branch. The input goes to a residual unit and the output goes to both branches:

Trunk branch: In this part there is two pre-activation Residual Unit sequentially.

Mask branch: In this branch, we have downsampling and upsampling parts. In the downsample part there are three blocks of Maxpooling and residual unit. After reaching the lowest resolution, the global information is then expanded by upsampling part to guide input features in each position. The upsampling part consists of three blocks of upsampling layer and a residual unit.

After the first and second blocks of downsampling, we have a skip connection to the second and last blocks of upsampling respectively. These skip connections are residual units.

The second attention module has one block and one skip connection less than the first attention module in the down-sampling and upsampling parts. The third module has the same relation with the second module.

The output of the mask branch $M(x)$ and trunk branch $T(x)$ should multiply element-wisely, But due to the range of the mask's output which is $[0,1]$, we will degrade the value of features in deep layers. to prevent this problem first we add 1 to the mask's output then do the multiply.

$$H(x) = (M(x) + 1) * T(x)$$

In the figure 5 this process is shown. $H(x)$ goes through a residual unit as the last step of the attention module.

Our implementation of the network follows the table shown in figure 6.

To train the model we use nesterov SGD optimizer and a weight decay of 0.0001 with a momentum of 0.9. we also set the initial learning rate to 0.0001. The learning rate is divided by 2 for every 30 epochs.

D. Training Process

To train all three models, we use batch size 32 and train them for 150 epochs. The objective of the training process is minimizing the categorical_crossentropy function with a specific optimizer for each model.

we use some callbacks in our implementations. First EarlyStopping is defined for checking the performance of the network on validation accuracy with patience = 50 for DenseNet and CovidNet and with patience = 30 for Attention ResNet. The other callback is ModelCheckpoint for saving the model with the best accuracy on the validation dataset to apply to the test dataset. Finally, we use ReduceLROnPlateau for the DenseNet and CovidNet, LearningRateScheduler for Attention ResNet, to update the learning rate (the detail is mentioned in each part).

E. weighted ensemble

In this step, we have the result of each model on the test dataset. Therefore as the output of the Sigmoid function, for each image, we have three probabilities related to each class. We also have the best validation accuracy for each model. probabilities and accuracies are the inputs for the ensemble.

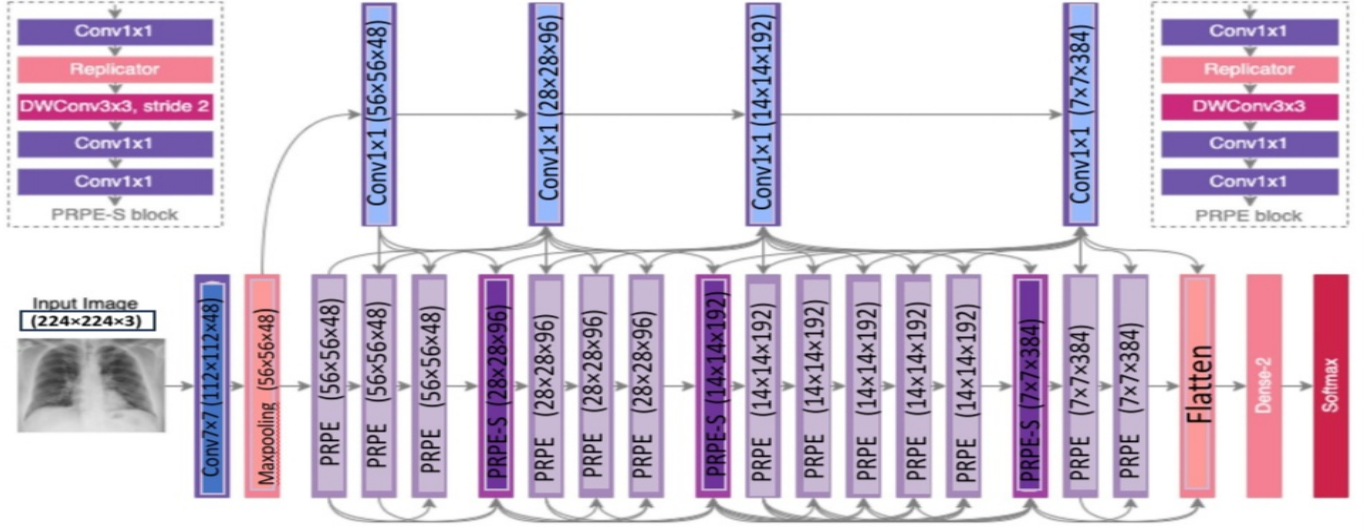


Fig. 4: The CovidNet architecture

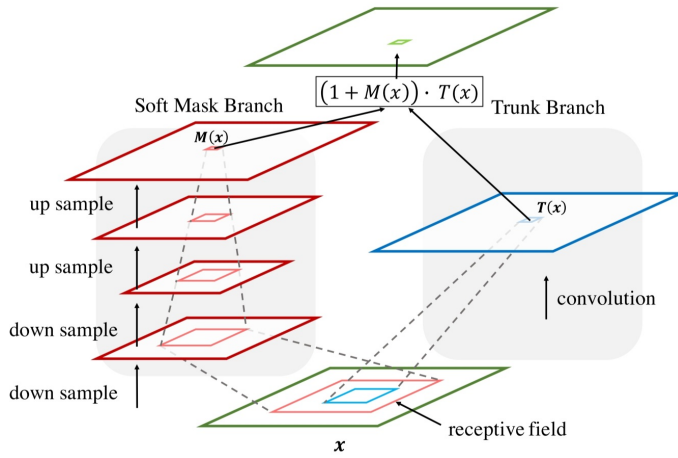


Fig. 5: The combination of trunk and mask output

For calculating the weight of each model, first we should have the minimum accuracy between models, then we subtract this minimum from all models' accuracies. These two steps follow these formulas:

$$\beta = \min_{j=1}^3 (accuracy_j)$$

$$Weight_j = accuracy_j - \beta$$

Where $j=1,2,3$ shows the models' accuracy. Then for each image, we multiply the weight to the probabilities and sum over three models:

$$Image_j = \operatorname{argmax}_i [\sum_{j=1}^3 weight_j * prob_{ji}]$$

Where $i=1,2,3$ shows the classes.

VI. RESULTS

In this part, we study the results of four implementations: DenseNet, CovidNet, Attention ResNet, and another version of Attention ResNet. In the new version of Attention ResNet,

Layer	Output Size	Attention
Conv1	112×112	7 × 7, 64, stride 2
Max pooling	56×56	3 × 3 stride 2
Residual Unit	56×56	$\begin{pmatrix} 1 \times 1, 32 \\ 3 \times 3, 32 \\ 1 \times 1, 128 \end{pmatrix} \times 1$
Attention Module	56×56	Attention × 1
Residual Unit	28×28	$\begin{pmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{pmatrix} \times 1$
Attention Module	28×28	Attention × 1
Residual Unit	14×14	$\begin{pmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{pmatrix} \times 1$
Attention Module	14×14	Attention × 1
Residual Unit	7×7	$\begin{pmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{pmatrix} \times 3$
Average pooling	1×1	7 × 7 stride 1
FC, Softmax		

Fig. 6: The Attention ResNet architecture

we use the same structure, just in Residual units we use stride=1 for all convolution layers. the result of this new network has an improvement in the test accuracy by 2.19% in comparison with the first Attention ResNet. In the ensemble model, we just choose the three best models as inputs of ensemble learning.

The comparison between the loss and accuracy of the training process for all networks is shown in plots in figure 7 and 8.

In the accuracy plot, we can see that the first Attention ResNet model is stopped at epoch 118 since it was not able to

	Attention ResNet'1	Attention ResNet'2	CovidNet	DenseNet	Ensemble
Training Time (minutes)	202.03	235.79	226.71	217.35	-
learnable parameters	8597123	6007203	4690323	6965001	-
Test Accuracy (percentage)	89.29	91.48	94.1	95.74	95.85
recall(percentage)	89	91	94	96	96
Precision(percentage)	89	91	94	96	96
F1-score (percentage)	89	91	94	96	96
F1-score Normal (percentage)	85	88	92	94	94
F1-score Covid (percentage)	91	94	96	99	99
F1-score pneumonia (percentage)	92	93	94	94	94

TABLE 1: Models' comparison

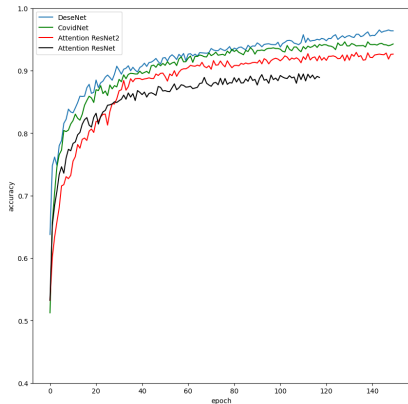


Fig. 7: The train accuracy of all models

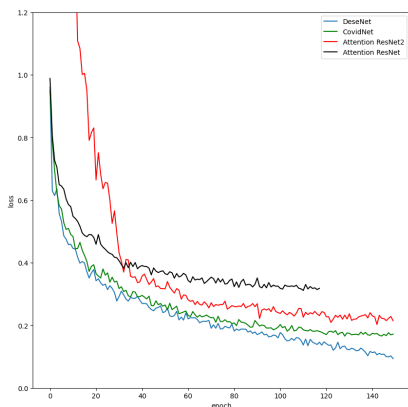


Fig. 8: The train loss of all models

get better in validation accuracy. Totally it is shown that the DenseNet model has the better performance among all the models. In the loss plot, the second Attention ResNet starts from a high value for loss but it decreases faster than the first Attention ResNet. The first Attention ResNet converges very slowly and there was no improvement in the last epochs. It can be seen also the CovidNet reached a stable level and does not improve anymore. However, we can see that the DenseNet's trend at the last epoch is still decreasing, and if it has more time it could perform better results.

In the figure 9, we can see that Ensemble, DenseNet, and CovidNet models have good performance in classifying

Normal images and the last four heatmap shows good classifying in covid images. But all models have problems in recognizing pneumonia and mostly they put it in the normal class which gives us low sensitivity. In the ensemble model's heatmap, we can see a slight improvement in the result in comparison with DenseNet. It classified 1 normal image truly while the DenseNet recognize it as pneumonia.

In the table, 1 model comparison is shown. we can see in this table that totally the ensemble and DenseNet have the best performances. The ensemble can increase the DenseNet accuracy by less than 1%. DenseNet network has the second less training time with the highest accuracy which is the best mode among all networks. The number of parameters in CovidNet is less than the others.

VII. CONCLUSION

This paper besides investigating the three models' performances, tried to use a weighted ensemble to increase the overall accuracy. We can see that the dense connections in the DenseNet and CovidNet play a crucial role in image classification since there is a good balance between the training time, learnable parameters, and accuracy. The DenseNet has also acceptable complexity.

For Attention ResNet we need more data and more epochs to achieve comparable results that in our case we do not have enough sources to continue the training process.

A remarkable result of our study is that all of our models failed to classify pneumonia images and predict them as normal images. It needs more data and deeper study in data augmentation for solving this problem.

During the project, we face difficulty in regulating the decay of the learning rate during the training. We try different strategies to find the proper setting.

Utilizing the ensemble model increased our accuracy slightly and this model can be more effective when we have the networks (as ensemble inputs) that cover each other's weaknesses. When all have weaknesses in one part like the problem in classifying pneumonia images, the ensemble can not increase the accuracy significantly.

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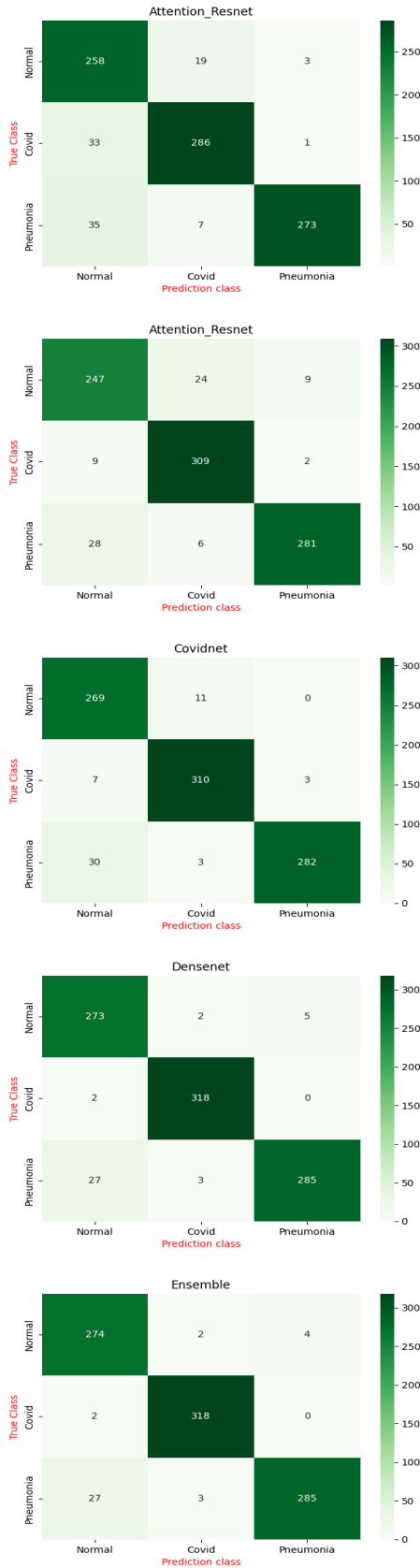


Fig. 9: The heatmap of the confusion matrix for Attention ResNet_1(First), Attention ResNet_2 (Second), Covid-Net(Third), DenseNet (Fourth), Ensemble (fifth)